Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational Scientist (PhD, Physics) with 10+ years building high-performance numerical algorithms and physics-based simulations. Expertise in scientific machine learning and geospatial data analysis. Architect of open-source simulation libraries, leader of funded research projects, and translator of complex physical models into production-quality scientific software across cross-functional research and engineering teams.

Skills

Languages: Python, C/C++, R, Rust, SQL, Fortran 95, Java, Julia, Shell
ML & Data: Scientific machine learning, PyTorch, scikit-learn, TensorFlow, random forest, neural networks, SVM, gradient-based optimization, ML-assisted parameter estimation, NumPy, SciPy, Pandas, Astropy
Geospatial: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS
HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5
Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, ParaView, VisIt

Experience

Independent Contractor, *Mercor* Jan 2026 – Present
Physics subject-matter expert evaluating frontier language models for multiple leading AI labs.

- Review 50+ expert-level physics, hydrodynamics, astrophysics, and cosmology problems for leading AI labs; write original prompts that stump frontier models and diagnose failure modes in their reasoning.

Spatial Data Scientist, *TealWaters* May 2025 – Jan 2026
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Independent Researcher & Open-Source Developer, *Self-directed* Jan 2024 – May 2025
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer to model radiation from relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.

Software Engineer, *Paychex* Apr 2022 – Jan 2024
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate, *Rochester Institute of Technology* Feb 2021 – Apr 2022
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries; added black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo against Fermi-LAT gamma-ray spectra with gradient-based optimization that updates the model automatically; supervised a graduate student on the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers.

Postdoctoral Research Fellow, *Purdue University* Oct 2018 – Nov 2020
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer; optimized it with OpenMP and tested GPU offload with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).

- Served as co-I on two NASA Fermi Guest Investigator grant proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* Jan 2018 – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant, *Universitat de València* Oct 2011 – Jul 2017

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* Jul 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* Sep 2011

Universidad Autónoma del Estado de México, *B.Sc. in Physics* Dec 2008

Publications

Articles (9 published, 3 first-author; 3 in preparation)

[12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C. *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.

[11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C. *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.

[10] **Rueda-Becerril, J. M.**, Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.

[9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.

[8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.

[7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D 106, 083015 (2022), arXiv:2112.09817.

[6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ 919, 95 (2021), arXiv:2106.05356.

[5] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.

[4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.**, & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS 496, 974–986 (2020), arXiv:1910.14049.

[3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

[2] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS 438, 1856–1869 (2014), arXiv:1310.5441.

[1] Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.

Conference Proceedings

[P5] **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr. 342, 277–282 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13797.

[P4] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *The blazar sequence revised*, Astron. Nachr. 342, 147–152 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13805.

[P3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) 233, 159 (2014), arXiv:1502.07882.

[P2] **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A., & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. 61, 02007 (2013), arXiv:1309.4612.

[P1] Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S., & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser. 454, 012001 (2013), arXiv:1211.1794.

Grants and Fellowships

Research Grants

NASA Fermi Guest Investigator Program, Cycle 12 2019
A simple model to understand the blazar sequence Grant #121077, $68K
PI: D. Giannios; Co-I: J. M. Rueda-Becerril

Fellowships

International Postdoctoral Fellowship Oct 2018 – Nov 2020
National Council of Science and Technology (CONACyT), Mexico

Postdoctoral Fellowship Jan 2018 – Sep 2018
Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)

Study-Abroad Fellowship Sep 2014 – Aug 2016
National Council of Science and Technology (CONACyT), Mexico

Santiago Grisolía Fellowship Oct 2011 – Jun 2014
Council of Education, Research, Culture and Sport, Valencian Community, Spain

M.Sc. Fellowship Sep 2009 – Aug 2011
CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo

Summer Research Fellowship (XVII Summer of Scientific Research) Jun 2007 – Aug 2007
Mexican Academy of Sciences

Projects

Blazar Population ML Jan 2026 – Present
Scientific ML (Python, scikit-learn, PyTorch) github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, to be validated against Tleco output (in progress).
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

Awards

Marcos Moshinsky Award for Best Poster Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference

Lic. Juan Josafat Pichardo Cruz Award 2009
Universidad Autónoma del Estado de México
For defending a licentiate thesis within a year of completing the undergraduate coursework.

Presentations

Invited Talks

The Role of Machine Learning in Environmental Science Mar 2024
Seminar on Climate Change Ecology, University of Washington Seattle, WA

Morphology of the spectra from numerical simulations of the internal shocks model for blazars Feb 2019
Astrophysics Seminar, Purdue University West Lafayette, IN

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars Jun 2018
DATA Group Weekly Seminar, Instituto de Astronomía, UNAM Mexico City, Mexico

Numerical treatment of non-thermal radiation in the internal shocks model for blazars Mar 2018
Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Oct 2014
IVICFA's Fridays: Computation in Physics, IFIC Paterna, Spain

Contributed Talks

Simulations of supermassive binary black holes accretion dynamics in the spinning case Apr 2022
APS April Meeting 2022 New York, NY

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows Apr 2021
APS April Meeting 2021 Online

The blazar sequence revised Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
Video: youtube.com/watch?v=BAZNWLNT69M

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario Jun 2014
Extreme-Astrophysics in an Ever-Changing Universe Ierápetra, Greece

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jul 2013
XXXIV Biennial Meeting of the Royal Spanish Society of Physics Valencia, Spain

Posters

A numerical approach for radiative cooling in relativistic outflows Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPI

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Dec 2014
Swift: 10 Years of Discovery Rome, Italy

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jun 2013
The Innermost Regions of Relativistic Jets and Their Magnetic Fields Granada, Spain

Teaching and Mentoring

Zachary Davis, graduate student 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored papers [8, 9]

Amanda O. Harrison, graduate student 2018 – 2020
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored paper [5]

Hao Zhang, graduate student 2018 – 2019
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored paper [4]

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Professional Development

High Performance Computing on Frontera May – Jun 2021
Jason Allison et al., TACC, Austin, TX Lecture series

Writing Winning Grants Nov 2019
Dr. Lauren Broyles, Purdue University Lecture

XSEDE HPC Workshop: Summer Boot Camp Jun 2019
John Urbanic, Purdue University Workshop

Data Analysis and Machine Learning with Python Feb 2017
Dr. Alejandro Torres, Universitat de València Workshop

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit Jul 2016
Dr. Vassilios Mewes, Universitat de València Workshop, 8 hours

The Universe in the Light of PLANCK and BICEP2 May 2014
Prof. Nick Mavromatos, Universitat de València Lecture series, 2 credits

Dark Matter Sep 2013
Prof. Alejandro Ibarra, Universitat de València Lecture series, 2 credits

International Cargèse School on Cosmic Accelerators Apr – May 2013
Institut d'Études Scientifiques de Cargèse, France Summer school

Introduction to C++ Programming Apr 2012
Dr. Jacek Generowicz, Universitat de València Workshop, 6 credits

Numerical Relativistic Astrophysics Mar – Apr 2012
Prof. Luciano Rezzolla, Universitat de València Lecture series, 9 hours

Fortran for Scientific Computing Mar 2012
HLRS, University of Stuttgart, Germany Workshop, 33 hours

Outreach

Los más rápidos y los más furiosos (The Fastest and the Most Furious) Sep 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) Aug 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes) May 2020
Científicos Mexicanos en el Extranjero Blog post
mexiciencia.github.io/post/anita

¿Qué es el modelo SIR? (What is the SIR Model?) May 2020
Científicos Mexicanos en el Extranjero Blog post
mexiciencia.github.io/post/modelo-sir

Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak) Apr 2020
Científicos Mexicanos en el Extranjero Blog post
Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19

Annual Department of Physics and Astronomy Poster Event Nov 2019
Purdue University 3 posters

Post-Doc Panel Q&A: What Happens When We Complete Our PhDs? Apr 2019
Department of Physics and Astronomy, Purdue University Panelist

Annual Department of Physics and Astronomy Poster Event Nov 2018
Purdue University Poster

¿Decía Einstein la verdad? (Was Einstein Telling the Truth?) Mar 2009
Facultad de Ciencias, Universidad Autónoma del Estado de México Talk

Service

Organizing contributor Dec 2012
X Scientific Meeting of the Spanish Astronomical Society Valencia, Spain

Student representative, Governing Council of the Faculty of Sciences Aug 2007 – May 2009
Universidad Autónoma del Estado de México Toluca, Mexico

Languages

Spanish (native), English (full professional proficiency), Catalan (basic), Nahuatl (basic)